Expanding Emotet Dataset –
[Malware-Traffic-Analysis.net - 2023-03-16 - Epoch 5 activity]

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1. Dataset Selection

- Identified additional Emotet network traffic from Malware Traffic Analysis (2023)
- Selected a capture containing:
  - ➔ Emotet Epoch 5 infection activity
  - ➔ Associated spambot/network behaviour
- Verified suitability by reviewing the provided activity description (activity.txt):
  - ➔ Confirmed full infection chain presence
  - ➔ Ensured post-infection network behaviour (aligned with project scope)

2. Data Acquisition

- Downloaded the corresponding PCAP file from the dataset
- Stored it within the project directory: data/emotet/pcap/
- Maintained consistent file organisation to align with existing pipeline structure

3. Network Log Extraction (Zeek Processing)

- Processed the PCAP file using Zeek: Converted raw packet data into structured logs
- Generated key log files, including:
  - ➔ conn.log (primary for behavioural features)
  - ➔ dns.log, http.log, smtp.log, ssl.log (supporting context)
- Stored outputs in: data/emotet/zeek_logs/mta_2023_03_16/

4. Infected Host Identification

- Analysed conn.log to determine the compromised machine
- Identified the infected host based on:
  - ➔ High volume of outbound connections
  - ➔ Communication with multiple external IP addresses
  - ➔ Activity across multiple protocols and ports
  - ➔ Determined infected host IP as: 10.3.16.101

5. Integration into Feature Pipeline

- Added the new dataset to the Emotet feature engineering pipeline
- Registered it as a new log source in: build_emotet_windows.py
- Included:
  - ➔ Log file path
  - ➔ Identified infected host IP
  - ➔ Correct class label (Emotet = 2)

6. Purpose and Impact

- Increased diversity of Emotet behavioural patterns in the dataset
- Improved representation of:
  - ➔ Modern infection techniques
  - ➔ Real-world network behaviour
  - ➔ Strengthens model generalisation and robustness against unseen attacks

7. Behavioural Window Generation

- Executed the Emotet feature engineering pipeline to regenerate behavioural windows: python src/emotet/build_emotet_windows.py
- The pipeline processed all registered log sources, including the newly added dataset
- Generated an updated dataset: data/emotet/emotet_windows.csv
- Outcome:
  - ➔ Total rows: 5,011
  - ➔ Class distribution: Normal (0): 2,821, Emotet (2): 2,190
  - ➔ Verified successful integration by inspecting the dataset_id column:
- Confirmed presence of new dataset: mta_2023_03_16 → 1,635 windows
- Ensured:
  - ➔ No data corruption
  - ➔ All previous datasets remained intact
  - ➔ Feature schema unchanged

8. Unified Dataset Reconstruction:

- Rebuilt the unified multi-class dataset: python scripts/build_unified_multiclass.py
- Combined:
  - ➔ SQL injection feature dataset
  - ➔ Updated Emotet behavioural windows
- Output: data/unified_multiclass.csv
- Outcome:
  - ➔ Total rows: 35,930
- Class distribution:
  - ➔ Normal (0): 22,358
  - ➔ SQLi (1): 11,382
  - ➔ Emotet (2): 2,190
- Verified:
  - ➔ Correct class encoding (0, 1, 2)
  - ➔ Consistent feature set (26 features)
  - ➔ Successful merge without loss or duplication

9. Balanced Dataset Construction

- Updated class balancing strategy to reflect new Emotet count
- Set: TARGET_PER_CLASS = 2190
- Rebuilt balanced dataset: python scripts/build_unified_multiclass_balanced.py
- Output: data/unified_multiclass_balanced.csv
- Outcome:
  - ➔ Total rows: 6,570
- Class distribution:
  - ➔ Normal (0): 2,190
  - ➔ SQLi (1): 2,190
  - ➔ Emotet (2): 2,190
- Ensured:
  - ➔ Perfect class balance
  - ➔ Feature consistency maintained

10. Model Retraining — Unified Dataset

- Retrained the unified multi-class Random Forest model: python scripts/train_unified_multiclass.py
- Outcome:
  - ➔ Macro F1-score: 0.9866
- Class-wise F1:
  - ➔ Normal: 0.9924
  - ➔ SQLi: 0.9892
  - ➔ Emotet: 0.9782
- Cross-validation:
  - ➔ Mean Macro F1: 0.9879
  - ➔ Standard deviation: 0.0021
- Additional evaluation:
  - ➔ ROC-AUC and PR-AUC scores close to 1.0 across all classes
  - ➔ Random Forest significantly outperformed Logistic Regression

11. Model Retraining — Balanced Dataset

- Trained a second model using the balanced dataset: python scripts/train_unified_multiclass_balanced.py
- Outcome: Achieved perfect classification performance:
- Precision, Recall, F1-score = 1.00 for all classes
- Interpretation:
  - ➔ Demonstrates model capacity under balanced conditions
  - ➔ Used as a comparative experiment rather than a primary result

12. Final Validation and Environment Alignment

- Verified model integrity: Confirmed correct class labels and feature count (26 features)
- Ensured compatibility between:
  - ➔ Model artifact
  - ➔ Python environment
  - ➔ Scikit-learn version
- Standardised environment using: Python 3.11 virtual environment (.venv311)
- All training, evaluation, and model loading operations were conducted within this environment to ensure consistency and avoid version mismatch issues

13. Overall Impact of Dataset Expansion

- Increased Emotet samples from ~555 → 2,190
- Significantly improved minority class representation
- Enhanced model performance on Emotet detection:
- Strong F1-score with stable cross-validation
- Key outcome:
  - ➔ Improved generalisation to unseen behavioural patterns
  - ➔ Reduced bias toward majority classes
  - ➔ Strengthened validity of the unified detection approach

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Comparisons:

Metric	Before Expansion	After Expansion
Emotet Samples	~555	**2,190**
Total Dataset Size	~31,496	**35,930**
Emotet Proportion	~1.7%	**6.1%**
Feature Count	26	26 (unchanged)
Model Type	Random Forest	Random Forest (unchanged)

Metric	Before Expansion	After Expansion
Macro F1-score	~0.9913 (random split) / 0.8546 (group holdout)	**0.9866**
Emotet F1-score	~0.9259 (random split)	**0.9782**
Cross-validation Mean F1	~0.970–0.991	**0.9879**
Cross-validation Std	~0.001–0.005	**0.0021**

The expansion of the Emotet dataset resulted in a substantial improvement in minority class performance without altering the model architecture or feature set. This demonstrates that the original limitation was primarily due to insufficient behavioural diversity rather than model capacity. By incorporating additional real-world traffic from a separate capture, the model was exposed to a wider range of Emotet activity patterns, leading to improved generalisation and more reliable detection of previously underrepresented behaviours.

Put these in

Post-expansion group holdout results:

Metric	Value
F1 Normal (held out)	0.9191
F1 Emotet (held out)	0.9315
Weighted F1	0.9262
Macro F1	0.6169 — exclude from reporting, SQLi support = 0

Pre vs post expansion comparison (Emotet F1):

Split	Pre-expansion	Post-expansion
Random 80/20	0.926	0.978
Group Holdout	0.739	0.9315

The group holdout Emotet F1 improved from 0.739 to 0.9315 — that's the headline result from the dataset expansion. The model now generalises substantially better to unseen Emotet captures.